DS HW 3

1.Write a C program to store n elements in an array using dynamic memory allocation and find the largest number.

```
#include <stdio.h>

 #include <stdlib.h>

int main() {

        int n, i, largest; int *a;

        printf("Enter number of elements: ");
        scanf("%d", &n);

        a = (int *)malloc(n * sizeof(int));

        printf("Enter %d elements:\n", n);
       for(i = 0; i < n; i++)
                scanf("%d", &a[i]);

        largest = a[0];

        for(i = 1; i < n; i++)
        {
                if(a[i] > largest)
                        largest = a[i];
        }

        printf("Largest number = %d", largest);

        free(a);

        return 0;

}
```

Output:

Enter number of elements: 5

Enter 5 elements:

12 45 23 67 34

Largest number = 67

2. Reverse an Array Using Pointers

```
#include <stdio.h>

int main()
{
    int a[5], i;
    int *p;

    printf("Enter 5 elements:\n");
    for(i = 0; i < 5; i++)
        scanf("%d", &a[i]);

    p = a + 4;

    printf("Array in reverse order:\n");

    for(i = 0; i < 5; i++)
    {
        printf("%d ", *p);
        p--;
    }

    return 0;
```

```
}
```

Output:

Enter 5 elements:

10 20 30 40 50

Array in reverse order:

50 40 30 20 10

3. Student Information Using Structure and Sort by Name:

```
#include <stdio.h>

 #include <string.h>

struct Student { int rollno; char name[30]; float mark; };

int main() { struct Student s[3], temp; int i, j;

printf("Enter details of 3 students:\n");

for(i = 0; i < 3; i++)
{
   printf("\nStudent %d\n", i + 1);

   printf("Roll No: ");
   scanf("%d", &s[i].rollno);

   printf("Name: ");
   scanf("%s", s[i].name);

   printf("Mark: ");
   scanf("%f", &s[i].mark);
}


for(i = 0; i < 2; i++)
{
   for(j = i + 1; j < 3; j++)
   {
```

```
        if(strcmp(s[i].name, s[j].name) > 0)
        {
            temp = s[i];
            s[i] = s[j];
            s[j] = temp;
        }
    }
}

printf("\nStudents sorted by Name:\n");

for(i = 0; i < 3; i++)
{
    printf("%d\t%s\t%.2f\n",
          s[i].rollno, s[i].name, s[i].mark);
}

return 0;


}
```

Output:

Enter details of 3 students:

Student 1

Roll No: 101

Name: Ravi

Mark: 85

Student 2

Roll No: 102

Name: Arun

Mark: 92

Student 3

Roll No: 103

Name: Bala

Mark: 78

Students sorted by Name:

102 Arun 92.00

103 Bala 78.00

101 Ravi 85.00

4. Add N Distances in feet and inches Using Structures

```
#include <stdio.h>

struct Distance { int feet; int inch; };

int main() { struct Distance d[10], sum = {0, 0}; int n, i;

printf("Enter number of distances: ");
scanf("%d", &n);

printf("Enter distances (feet inch):\n");

for(i = 0; i < n; i++)
{
   printf("Distance %d: ", i + 1);
   scanf("%d %d", &d[i].feet, &d[i].inch);

   sum.feet += d[i].feet;
   sum.inch += d[i].inch;
}

sum.feet += sum.inch / 12;
sum.inch = sum.inch % 12;

printf("Total Distance = %d feet %d inches",sum.feet, sum.inch);
```

```
return 0;

}
```

Output:

```
Enter number of distances: 3
Enter distances (feet inch):
Distance 1: 5 8
Distance 2: 3 10
Distance 3: 4 6
Total Distance = 14 feet 0 inches
```

5. Add Two Complex Numbers by Passing Structure to Function

```
#include <stdio.h>

struct Complex
{
   float real;
   float imag;
};

struct Complex add(struct Complex c1, struct Complex c2)
{
   struct Complex result;

   result.real = c1.real + c2.real;
   result.imag = c1.imag + c2.imag;
```

```
    return result;
}

int main()
{
    struct Complex c1, c2, result;

    printf("Enter first complex number:\n");
    printf("Real part: ");
    scanf("%f", &c1.real);
    printf("Imaginary part: ");
    scanf("%f", &c1.imag);

    printf("Enter second complex number:\n");
    printf("Real part: ");
    scanf("%f", &c2.real);
    printf("Imaginary part: ");
    scanf("%f", &c2.imag);

    result = add(c1, c2);

    printf("Sum = %.2f + %.2fi", result.real, result.imag);

    return 0;
}
```

Output

Enter first complex number:

Real part: 4

Imaginary part: 5

Enter second complex number:

Real part: 3

Imaginary part: 2

Sum = 7.00 + 7.00i

6. Inventory Management System Using Structures

```
#include <stdio.h>
struct Product
{
    int id;
    char name[30];
    int quantity;
    float price;
};

int main()
{
    struct Product p[3];
    int i;

    printf("Enter details of 3 products:\n");
```

```
for(i = 0; i < 3; i++)
{
    printf("\nProduct %d\n", i + 1);

    printf("Product ID: ");
    scanf("%d", &p[i].id);

    printf("Product Name: ");
    scanf("%s", p[i].name);

    printf("Quantity: ");
    scanf("%d", &p[i].quantity);

    printf("Price: ");
    scanf("%f", &p[i].price);
}

printf("\nInventory Details:\n");
printf("ID\tName\tQuantity\tPrice\n");

for(i = 0; i < 3; i++)
{
    printf("%d\t%s\t%d\t\t%.2f\n",
           p[i].id,
           p[i].name,
           p[i].quantity,
           p[i].price);
```

```
    }

    return 0;
}
```

Output:

```
Enter details of 3 products:

Product 1
Product ID: 101
Product Name: Pen
Quantity: 50
Price: 10

Product 2
Product ID: 102
Product Name: Book
Quantity: 20
Price: 50

Product 3
Product ID: 103
Product Name: Bag
Quantity: 10
Price: 500
```

Inventory Details:

ID	Name	Quantity	Price
101	Pen	50	10.00
102	Book	20	50.00
103	Bag	10	500.00

7. Decimal to Binary, Octal and Hexadecimal

```
#include <stdio.h>
int main()
{
   int n;

   printf("Enter a decimal number: ");
   scanf("%d", &n);

   printf("Binary = ");
   int b[32], i = 0, temp = n;

   while(temp > 0)
   {
      b[i] = temp % 2;
      temp = temp / 2;
      i++;
   }

   while(i > 0)
   {
```

```
        printf("%d", b[--i]);
    }

    printf("\nOctal = %o", n);
    printf("\nHexadecimal = %X", n);

    return 0;
}
```

Output:

```
Enter a decimal number: 25
Binary = 11001
Octal = 31
Hexadecimal = 19
```

8. Difference Between Two Time Periods:

```
#include <stdio.h>
struct Time
{
    int hour;
    int minute;
    int second;
};

int main()
{
    struct Time t1, t2, diff;

    printf("Enter first time (hour minute second): ");
```

```
    scanf("%d %d %d", &t1.hour, &t1.minute, &t1.second);

    printf("Enter second time (hour minute second): ");
    scanf("%d %d %d", &t2.hour, &t2.minute, &t2.second);

    if(t2.second < t1.second)
    {
        t2.second += 60;
        t2.minute--;
    }

    if(t2.minute < t1.minute)
    {
        t2.minute += 60;
        t2.hour--;
    }

    diff.second = t2.second - t1.second;
    diff.minute = t2.minute - t1.minute;
    diff.hour = t2.hour - t1.hour;
    printf("Time Difference = %d:%d:%d", diff.hour, diff.minute, diff.second);
    return 0;
}
```

Output:

Enter first time (hour minute second): 10 30 20

Enter second time (hour minute second): 12 45 50

Time Difference = 2:15:30

9. Employee Management System Using Structures:

```
#include <stdio.h>
struct Employee
{
    int id;
    char name[30];
    float salary;
};

int main()
{
    struct Employee e[3];
    int i;

    printf("Enter details of 3 employees:\n");

    for(i = 0; i < 3; i++)
    {
        printf("\nEmployee %d\n", i + 1);

        printf("Employee ID: ");
        scanf("%d", &e[i].id);

        printf("Name: ");
        scanf("%s", e[i].name);

        printf("Salary: ");
```

```
        scanf("%f", &e[i].salary);
    }

    printf("\nEmployee Details:\n");
    printf("ID\tName\tSalary\n");

    for(i = 0; i < 3; i++)
    {
        printf("%d\t%s\t%.2f\n",
                e[i].id,
                e[i].name,
                e[i].salary);
    }

    return 0;
}
```

Output:

Enter details of 3 employees:

Employee 1

Employee ID: 101

Name: Arun

Salary: 25000

Employee 2

Employee ID: 102

Name: Ravi

Salary: 30000

Employee 3

Employee ID: 103

Name: Priya

Salary: 28000

Employee Details:

ID Name Salary

101 Arun 25000.00

102 Ravi 30000.00

103 Priya 28000.00

10. Area and Perimeter of Rectangle and Triangle Using Function Pointers:

```
#include <stdio.h>
 float rectangleArea(float l, float b)
{
   return l * b;
}

float rectanglePerimeter(float l, float b)
{
   return 2 * (l + b);
}
```

```
float triangleArea(float b, float h)
{
   return (b * h) / 2;
}
float trianglePerimeter(float a, float b, float c)
{
   return a + b + c;
}
int main()
{
   float l, b, h;
   float a, tb, tc;
    float (*area)(float, float);
   float (*perimeter)(float, float);
    printf("Enter length and breadth of rectangle: ");
   scanf("%f %f", &l, &b);
   area = rectangleArea;
   perimeter = rectanglePerimeter;

   printf("Area of Rectangle = %.2f\n", area(l, b));
   printf("Perimeter of Rectangle = %.2f\n", perimeter(l, b));
   printf("\nEnter base and height of triangle: ");
   scanf("%f %f", &tb, &h);
```

```
    printf("Enter the other two sides of triangle: ");
    scanf("%f %f", &a, &tc);
    area = triangleArea;
    printf("Area of Triangle = %.2f\n", area(tb, h));
    float (*triPerimeter)(float, float, float);
    triPerimeter = trianglePerimeter;
    printf("Perimeter of Triangle = %.2f\n",triPerimeter(tb, a, tc));
    return 0;
}
```

Output:

```
Enter length and breadth of rectangle: 10 5
Area of Rectangle = 50.00
Perimeter of Rectangle = 30.00

Enter base and height of triangle: 6 4
Enter the other two sides of triangle: 5 5
Area of Triangle = 12.00
Perimeter of Triangle = 16.00
```